CRIMELENS • VIVA QUICK REVISION & MASTER CHEAT SHEET

CrimeLens High-Yield Viva Defense Guide

A high-density technical summary of core architectures, mathematical formulas, algorithms, empirical results, and anticipated professor viva defense questions for the CrimeLens Computer Engineering capstone project.

PROJECT EXECUTIVE SNAPSHOT

- Project Name: CrimeLens — Chicago Crime Analytics Workstation
- Core Domain: Data Warehousing and Data Mining (DWM) / Geospatial Analytics
- Dataset Census: 761,718 authentic reported incidents (2023: 263,868; 2024: 259,670; 2025: 238,180)
- Geocoding Quality: 754,250 valid coordinates (99.02%); 7,468 mapped to surrogate key 0
- Stack: PostgreSQL 18.4 (Star Schema) • Python 3.13 / FastAPI • React 18 / Vite • Power BI
- Algorithms: DBSCAN Spatial Clustering • K-Means Profile Segmentation • PCA 2D Projection • Apriori Rule Mining • Decision Tree Classification

1. The Kimball Star Schema Architecture

Table Name	Type	Row Count	Surrogate PK	Key Dimensional Role
fact_crime	Fact Table	761,718	crime_fact_key (bigint)	Central fact table storing individual reported criminal complaints.
dim_time	Dimension	26,305	time_key (YYYYMMDDHH)	13 temporal attributes: year, quarter, month, day, hour, is_weekend.
dim_area	Dimension	1,004	area_key (int)	Administrative boundary hierarchy: district (24), beat (277), ward, area.
dim_crime	Dimension	360	crime_key (int)	Statutory classification: primary_type (31 types), description, IUCR, FBI code.
dim_location	Dimension	279,869	location_key (int)	Spatial coordinates: block address, latitude, longitude (key 0 for unknown).
dim_premise	Dimension	143	premise_key (int)	Environmental location description (e.g. STREET, APARTMENT, SIDEWALK).
dim_status	Junk Dim	4	status_key (int)	Combines binary flags: arrest (True/False) and domestic (True/False).

2. Master Data Mining Algorithm & Formula Cheat Sheet

Algorithm	Discipline & Task	Key Mathematical Formulas	Empirical CrimeLens Result

DBSCAN	Unsupervised Spatial Clustering (Micro-level)	Haversine Spherical Distance: a = sin^2(dlat/2) + cos*cos*sin^2(dlon/2) d = 2 * R * atan2(sqrt(a), sqrt(1-a)) Params: eps = 500m, MinPts = 30	14 high-density street corridor hotspots; 35% sparse incidents rejected as noise.
K-Means	Unsupervised Profile Segmentation (Macro-level)	StandardScaler: z = (x - mu) / sigma Inertia = SUM ||x - mu_k||^2 Silhouette: s = (b - a) / max(a, b)	24 Districts clustered across 11 features ($K=4$, Inertia: 57.73, Silhouette: 0.3914).
PCA	Dimensionality Reduction & Visualization	Eigen decomposition: Sigma * v = lambda * v Explained Variance: EVR = lambda_k / SUM(lambda)	2D scatterplot captures 72.39% of total variance across 11 features (PC1: 46.5%, PC2: 25.9%).
Apriori	Unsupervised Transaction Rule Mining	Support: P(A ∩ B) = Count(A ∪ B) / N Confidence: P(B | A) = Sup(A ∩ B) / Sup(A) Lift = Conf(A => B) / Sup(B)	Rule #68: {Motor Vehicle Theft} => {Arrest=NO, Street} (Sup: 6.49%, Conf: 72.4%, Lift: 3.09x).
Decision Tree	Supervised Classification (Arrest Outcome)	Gini Impurity: Gini = 1 - SUM(p_k^2) Class Weight: w = N / (C * N_c) ROC-AUC = Area Under TPR vs FPR Curve	Test Accuracy: 88.20% (beats 83.43% majority baseline); ROC-AUC: 0.7178; Overfitting Gap: -0.02%.

3. Confusion Matrix & Machine Learning Evaluation Metrics

• Accuracy = (TP + TN) / (TP + TN + FP + FN) — Dangerous alone in imbalanced datasets!
• Precision = TP / (TP + FP) — 'When model predicts arrest, how often is it right?' (CrimeLens: 89.39%)
• Recall = TP / (TP + FN) — 'Of all actual arrests, how many did the model catch?' (CrimeLens: 31.4% on minority arrest)
• F1-Score = 2 * (Precision * Recall) / (Precision + Recall) — Harmonic mean balancing FP and FN.
• ROC-AUC = Area under ROC curve across all thresholds (CrimeLens: 0.7178).

4. Top 20 Rapid-Fire Viva Questions & Winning Examiner Defenses

Q1: Why did you use PostgreSQL instead of MongoDB? A: Data warehousing relies on dimensional star schemas, relational integrity, integer surrogate key joins, and SQL OLAP aggregations. MongoDB lacks ACID multi-table join performance for analytical aggregation over 760k rows.

Q2: Why a Star Schema instead of 3NF relational normalization? A: 3NF requires dozens of complex table joins to eliminate redundancy, slowing down analytical read queries. A Star Schema denormalizes dimensions around a central fact table, allowing fast 1-hop star-joins.

Q3: Why DBSCAN instead of K-Means for crime hotspots? A: K-Means forces every incident into K spherical clusters and cannot handle noise. DBSCAN discovers arbitrary geometric shapes (like crime along curved streets) and filters out sparse background crime as noise.

Q4: Why K-Means for district segmentation if DBSCAN is better for hotspots? A: Different tasks! DBSCAN clusters granular micro-coordinates (lat, lon). K-Means segments aggregated municipal jurisdictions (24 police districts) across 11 standardized multivariate feature dimensions.

Q5: What is the significance of the 2D PCA variance of 72.39%? A: It proves that our 2-dimensional scatterplot successfully captures nearly three-quarters of all mathematical information and variance present across the original 11 feature dimensions without severe distortion.

Q6: Why can Silhouette recommend K=2 while you select K=4 in K-Means? A: Silhouette evaluates purely mathematical cluster tightness (where K=2 yields two broad groups: commercial vs residential). K=4 provides four nuanced operational typologies needed by law enforcement commanders.

Q7: What does Lift > 1.0 mean in Apriori association rules? A: Lift measures how much more frequently antecedent A and consequent B co-occur than expected if they were statistically independent. Lift = 3.09x means they appear together over three times more often than random chance.

Q8: Does an association rule prove that parking on a street causes auto theft? A: Absolutely not! Association does NOT imply causation. It proves an environmental vulnerability and statistical co-occurrence pattern in historical police logs.

Q9: Why is 88.20% accuracy in your Decision Tree actually meaningful? A: Because our majority-class baseline is 83.43% (non-arrests). An accuracy of 88.20% with balanced class weighting beats the baseline and achieves an ROC-AUC of 0.7178 with zero overfitting (-0.02% gap).

Q10: Why did you choose Import Mode instead of DirectQuery in Power BI? A: Import Mode loads data into the columnar VertiPaq engine, providing sub-second aggregations, full DAX time-intelligence function support (SAMEPERIODLASTYEAR), and offline report portability.

Q11: What is the Modifiable Areal Unit Problem (MAUP)? A: Statistical distortion occurring when point data (crimes) is aggregated into arbitrary administrative boundaries. Clustering 24 districts yields different patterns than clustering 77 community areas.

Q12: What is the Ecological Fallacy? A: The logical fallacy of assuming that characteristics of an aggregate district cluster apply to individual residents living within that district.

Q13: How do you know your 761,718 dataset is authentic? A: Direct live API reconciliation queries against the official City of Chicago Open Data Portal endpoint (data.cityofchicago.org/resource/ijzp-q8t2.json) confirmed a delta of exactly 0 across all three years.

Q14: What is the purpose of dim_status as a Junk Dimension? A: It combines the low-cardinality flags arrest (T/F) and domestic (T/F) into a single 4-row dimension, avoiding wide, sparse boolean columns in the fact table.

Q15: What is the difference between Slicing and Dicing? A: Slicing filters along exactly 1 dimension (creating a 2D slice); Dicing filters across 2 or more dimensions simultaneously (extracting a sub-cube).

Q16: How do you handle division by zero in DAX? A: We use the native DIVIDE() function with a safe third parameter: DIVIDE(Numerator, Denominator, 0).

Q17: What does WCSS stand for in K-Means? A: Within-Cluster Sum of Squares (Inertia). It measures the sum of squared Euclidean distances between points and their assigned centroids.

Q18: What is the Downward Closure Property in Apriori? A: If an itemset is frequent, all of its subsets must be frequent. If an itemset is infrequent, all of its supersets are pruned immediately.

Q19: What is the primary split in your Decision Tree? A: Primary crime classification: proactive enforcement crimes (narcotics, weapons) split toward arrest; stealth property crimes (theft, auto theft) split toward non-arrest.

Q20: What is the biggest limitation of CrimeLens? A: It analyzes closed historical census data (2023–2025) rather than streaming live 911 dispatch radio feeds.